

Topics to be covered

- Sequences, Bounded sequences, Monotonic Sequences, Limits of Sequences
- Limits of functions, Continuity of functions, Differentiability
- Intermediate Value Theorem, Rolle's theorem, Lagrange's MVT, Taylor's theorem, Darboux's theorem
- Critical points, local extrema (minima/maxima), global extrema (minima/maxima)
- Monotonicity of functions, concavity/convexity of functions, Point of inflection

A **Sequence** can be thought of as a function from $\mathbb{N}$ to some domain X .

For sequences of real numbers, we can write,

$$a : \mathbb{N} \rightarrow \mathbb{R}$$

We use the notation $\{a_n\}$ to represent a sequence, and a_n to represent the n^{th} term of the sequence,

$$a_1, a_2, a_3, \dots$$

Bounded Sequences

The sequence $\{a_n\}$ is called **bounded above**, if there exists an $M \in \mathbb{R}$ such that $a_n \leq M \forall n \in \mathbb{N}$. Then M is called an **upper bound** of $\{a_n\}$.

The sequence $\{a_n\}$ is called **bounded below**, if there exists an $m \in \mathbb{R}$ such that $m \leq a_n \forall n \in \mathbb{N}$. Then m is called a **lower bound** of $\{a_n\}$.

The sequence is called **bounded** if it is both bounded above and below.

For a bounded above sequence, M is called the **lowest upper bound**, if it is an upper bound of the sequence, and no number smaller than M is an upper bound of the sequence.

Similarly, define the **greatest upper bound** of a bounded below sequence.

Monotonic Sequences

A sequence $\{a_n\}$ in $\mathbb{R}$ is said to be **monotonically increasing** if $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}$.

A sequence $\{a_n\}$ in $\mathbb{R}$ is said to be **monotonically decreasing** if $a_n \geq a_{n+1}$ for all $n \in \mathbb{N}$.

A **monotonic sequence** is one that is either monotonically increasing or monotonically decreasing.

Examples of sequences

- $a_n = \frac{1}{n}$ is a decreasing, bounded sequence ($m = 0$ and $M = 1$)
- $a_n = \frac{n!}{n^n}$ is a decreasing, bounded sequence ($m = 0$ and $M = 1$)
- $a_n = n^{1/n}$ is a bounded sequence ($m = 0$ and $M = 3$)
(Note that 3 is *an* upper bound, but not *the least* upper bound!)
- $a_n = \sin(n)$ is a bounded sequence ($m = -1$ and $M = 1$)
- $a_n = n^2$ is increasing, bounded below, but not bounded above
- $a_n = n(-1)^n$ is neither bounded below, nor bounded above

Convergence using $\epsilon - N$ definition

$\epsilon - N$ Definition of Convergence

A sequence $\{a_n\}$ in $\mathbb{R}$ is said to **converge** if there exists $\ell \in \mathbb{R}$ such that *for every* $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|a_n - \ell| < \epsilon \quad \forall n > N.$$

The real number ℓ is called the limit of the sequence $\{a_n\}$.

We write it as $\lim_{n \rightarrow \infty} a_n = \ell$ or simply $a_n \rightarrow \ell$ as $n \rightarrow \infty$

If a sequence does not converge, it is said to **diverge**.

Basically, this means that if ℓ is the limit of $\{a_n\}$, then for *sufficiently large* n , the sequence becomes forever *confined* to the ϵ -neighbourhood of ℓ . Moreover, we can choose ϵ as small as we like, and this would still hold true.

Useful results on limits

Uniqueness - If a sequence converges, its limit is unique.

Limit Theorems - If $\{a_n\}$ and $\{b_n\}$ converge, then we have

- $\lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$
- $\lim_{n \rightarrow \infty} (a_n b_n) = (\lim_{n \rightarrow \infty} a_n) \cdot (\lim_{n \rightarrow \infty} b_n)$
- If $\lim_{n \rightarrow \infty} b_n \neq 0$, then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}$.

Sandwich Theorem -

Suppose $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n$. If $\{b_n\}$ is a sequence satisfying $a_n \leq b_n \leq c_n$ for all n , then b_n converges and

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} c_n.$$

Boundedness, Monotonicity and Convergence

Theorem (Boundedness of Convergent sequences)

If a sequence is $\mathbb{R}$ is convergent, then it is bounded.

This result can be used to prove that a sequence does not converge. If we can show that the sequence is not bounded, then we have proved that it does not converge.

Theorem (Convergence of Monotonic, Bounded sequences)

- A monotonically increasing sequence which is bounded above is convergent, and converges to its lowest upper bound.*
- A monotonically decreasing sequence which is bounded below is convergent, and converges to its greatest upper bound.*

Cauchy Sequences

A sequence $\{a_n\}$ in $\mathbb{R}$ is said to be **Cauchy** if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$, such that

$$|a_n - a_m| < \epsilon \quad \forall m, n \geq N.$$

This basically means, that for *sufficiently large* N , the elements of the sequence come within an ϵ -neighbourhood of each other.

Theorem

A sequence of real numbers is convergent if and only if it is a Cauchy sequence.

Exercise 1

Prove using $\epsilon - N$ definition, that

$$\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$$

Exercise 1 - Solution

Rationalize:

$$\sqrt{n+1} - \sqrt{n} = \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

Bound the denominator: since $\sqrt{n+1} > \sqrt{n}$,

$$0 < \sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{2\sqrt{n}}$$

ϵ -N argument: Let $\epsilon > 0$. Choose $N = \frac{1}{4\epsilon^2}$.

Then for all $n > N$,

$$\frac{1}{2\sqrt{n}} < \frac{1}{2\sqrt{N}} = \frac{1}{2 \cdot \frac{1}{2\epsilon}} = \epsilon$$

$$\therefore \left| \sqrt{n+1} - \sqrt{n} - 0 \right| = \sqrt{n+1} - \sqrt{n} < \frac{1}{2\sqrt{n}} < \epsilon \quad \forall n > N$$

Therefore $\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$.

Exercise 2

Prove that if $\lim_{n \rightarrow \infty} a_n$ exists, then $\lim_{n \rightarrow \infty} (a_{n+1} - a_n) = 0$.

Is the converse true? If so, prove it. If not, give a counterexample

Exercise 2 - Solution

Proof. Suppose $\lim_{n \rightarrow \infty} a_n = L$.

Since $\{a_{n+1}\}$ is just $\{a_n\}$ shifted by one index,

$$\lim_{n \rightarrow \infty} a_{n+1} = L \quad \text{as well.}$$

By the **limit theorem** (limit of a difference of sequences):

$$\lim_{n \rightarrow \infty} (a_{n+1} - a_n) = \lim_{n \rightarrow \infty} a_{n+1} - \lim_{n \rightarrow \infty} a_n = L - L = 0.$$

Converse: If $\lim(a_{n+1} - a_n) = 0$, does $\lim a_n$ exist?

False. Counterexample: $a_n = \sqrt{n}$.

$$a_{n+1} - a_n = \sqrt{n+1} - \sqrt{n} \rightarrow 0 \quad (\text{shown earlier}),$$

but $a_n = \sqrt{n} \rightarrow \infty$, so $\{a_n\}$ diverges.

Exercise 3

Let $\{a_n\}$ and $\{b_n\}$ be two sequences of real numbers. Let a and b be real numbers. The notation $a_n \not\rightarrow$ says that $\{a_n\}$ does not converge to a real number. Which one of the following statements is **FALSE**?

- (A) If $a_n \rightarrow a$, then $\{a_n\}$ is bounded.
- (B) If $a_n \rightarrow a$ and $b_n \rightarrow b$, then $a_n + b_n \rightarrow a + b$.
- (C) If $a_n \rightarrow 0$ and $\{b_n\}$ is bounded, then $a_n b_n \rightarrow 0$.
- (D) If $a_n \not\rightarrow$ and $b_n \not\rightarrow$, then $a_n + b_n \not\rightarrow$.

Exercise 3 - Solution

Part (C) Since $\{b_n\}$ is bounded, there exists $M > 0$ such that

$$|b_n| \leq M \quad \text{for all } n.$$

Then for all n ,

$$|a_n b_n - 0| = |a_n| |b_n| \leq M |a_n|.$$

ϵ -N argument: Let $\epsilon > 0$. Since $a_n \rightarrow 0$, applying the definition with the choice of neighbourhood to be $\frac{\epsilon}{M}$, there exists N such that

$$n > N \implies |a_n| < \frac{\epsilon}{M}.$$

Then for all $n > N$,

$$|a_n b_n - 0| \leq M |a_n| < M \cdot \frac{\epsilon}{M} = \epsilon.$$

Hence $a_n b_n \rightarrow 0$.

Exercise 3 - Solution

Part (D) The statement is false, a counterexample is -

$$a_n = n \quad \text{and} \quad b_n = -n$$

Then a_n and b_n diverge, but their sum $a_n + b_n = 0$ converges trivially

Exercise 4

Let $\{a_n\}$ be the sequence defined by

$$a_n = \sum_{k=1}^n \frac{1}{\sqrt{n^2 + k}}.$$

Then find $\lim_{n \rightarrow \infty} a_n$

Exercise 4 - Solution

For each $k = 1, \dots, n$:

$$\frac{1}{\sqrt{n^2 + n}} \leq \frac{1}{\sqrt{n^2 + k}} \leq \frac{1}{\sqrt{n^2}}$$

Summing over $k = 1$ to n :

$$\sum_{k=1}^n \frac{1}{\sqrt{n^2 + n}} \leq \sum_{k=1}^n \frac{1}{\sqrt{n^2 + k}} \leq \sum_{k=1}^n \frac{1}{\sqrt{n^2}}$$

$$\frac{n}{\sqrt{n^2 + n}} \leq a_n \leq \frac{n}{n}$$

$$\frac{1}{\sqrt{1 + \frac{1}{n}}} \leq a_n \leq 1$$

As $n \rightarrow \infty$, both bounds $\rightarrow 1$. Then apply sandwich theorem to get the limit.

Exercise 5

Prove that the sequence $a_n = c^{1/n}$, for some $c > 1$, converges, and find its limit.

(Hint - Apply AM-GM inequality for the set of numbers $\{c, 1, 1, \dots, 1\}$, i.e., one c and $(n - 1)$ 1s. Then use sandwich theorem)

Exercise 6

Does the sequence $a_n = (2^n + 3^n)^{1/n}$ converge? If so, find its limit.

Exercise 6 - Solution

Bound: Since $3^n \leq 2^n + 3^n \leq 3^n + 3^n = 2 \cdot 3^n$,

$$3^n \leq 2^n + 3^n \leq 2 \cdot 3^n$$

Taking n -th roots (increasing function on positives):

$$3 \leq (2^n + 3^n)^{1/n} \leq 2^{1/n} \cdot 3$$

Take limits: As $n \rightarrow \infty$, $2^{1/n} \rightarrow 1$ (given), so

$$\lim_{n \rightarrow \infty} 3 = 3, \quad \lim_{n \rightarrow \infty} 2^{1/n} \cdot 3 = 1 \cdot 3 = 3$$

By the **sandwich theorem**,

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (2^n + 3^n)^{1/n} = 3.$$

Limits of Functions (using Sequences)

Consider a function $f : D \rightarrow \mathbb{R}$, defined on a domain $D \subset \mathbb{R}$

Limits of Functions using sequences

If for every sequence $\{x_n\}$ in D , with $x_n \neq c$ and

$$\lim_{n \rightarrow \infty} x_n = c \quad \text{implies} \quad \lim_{n \rightarrow \infty} f(x_n) = L,$$

then we say that the **limit** of the function f at $x = c$ is L .

We write it as $\lim_{x \rightarrow c} f(x) = L$.

This definition is useful in proving that a function's limit does not exist. If you can give an example of two sequences $\{x_n\}$ and $\{y_n\}$, both converging to c , but if $\lim_{n \rightarrow \infty} f(x_n) \neq \lim_{n \rightarrow \infty} f(y_n)$, then that proves that the function's limit does not exist.

Limits of Functions (using ϵ - δ definition)

$\epsilon - \delta$ definition of Limit of a function

Iff for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - L| < \epsilon \quad \forall x \in (c - \delta, c) \cup (c, c + \delta),$$

then the number L is the **limit** of $f(x)$ at $x = c$.

We write it as $\lim_{x \rightarrow c} f(x) = L$.

This is an equivalent way to define limit of a function.

Right Hand and Left Hand Limits

We can define right-hand and left-hand limits using sequences -

If for every sequence $\{x_n\}$ in the domain of f ,

$$x_n \rightarrow c \text{ and } x_n > c \text{ implies } f(x_n) \rightarrow \ell_1,$$

then we call ℓ_1 the **right hand limit** of $f(x)$ at $x = c$, and write it as

$$\lim_{x \rightarrow c^+} f(x) = \ell_1$$

Similarly, if for every sequence $\{x_n\}$ in the domain of f ,

$$x_n \rightarrow c \text{ and } x_n < c \text{ implies } f(x_n) \rightarrow \ell_2,$$

then we call ℓ_2 the **left hand limit** of $f(x)$ at $x = c$, and write it as

$$\lim_{x \rightarrow c^-} f(x) = \ell_2$$

Right Hand and Left Hand Limits

Theorem

$\lim_{x \rightarrow c} f(x)$ exists if and only if $\lim_{x \rightarrow c^-} f(x)$ and $\lim_{x \rightarrow c^+} f(x)$ exist and are equal. In this case,

$$\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c} f(x).$$

Continuity of a Function

Continuity of a function

A function $f(x)$ is said to be continuous at $x = c$, if $\lim_{x \rightarrow c} f(x)$ exists and $\lim_{x \rightarrow c} f(x) = f(c)$

f is said to be continuous on D if f is continuous at each point of D .

f is said to be discontinuous on D if it is not continuous on D , i.e., there exists $c \in D$ such that f is not continuous at c .

Properties of Continuous Functions

If f and g are continuous at $x = c$, then -

- $f \pm g$ is also continuous at $x = c$
- fg is also continuous at $x = c$
- f/g is also continuous at $x = c$, provided $g(c) \neq 0$

Continuity of Composite functions -

If f is continuous at c , and g is continuous at $f(c)$, then $(g \circ f)(x) = g(f(x))$ is continuous at $x = c$

Differentiability of a function

A function f is said to be **differentiable** at $x = c$, if

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \text{ exists}$$

In this case, this limit is denoted by $f'(c)$, and it is called the derivative of f at $x = c$.

The left hand limit and right hand limit of this, are called respectively, the **left hand derivative** and **right hand derivative** of f at $x = c$.

Theorem

If f is differentiable at $x = c$, then it is continuous at $x = c$.

Exercise 7

Prove that the limit of a function is unique using $\epsilon - \delta$ definition.

Exercise 7 - Solution

Proof. Suppose, for contradiction, that $L_1 \neq L_2$.

Let $\epsilon = \frac{|L_1 - L_2|}{2} > 0$.

Since $\lim_{x \rightarrow a} f(x) = L_1$, there exists $\delta_1 > 0$ such that

$$0 < |x - a| < \delta_1 \implies |f(x) - L_1| < \epsilon$$

Since $\lim_{x \rightarrow a} f(x) = L_2$, there exists $\delta_2 > 0$ such that

$$0 < |x - a| < \delta_2 \implies |f(x) - L_2| < \epsilon$$

Let $\delta = \min(\delta_1, \delta_2)$. Pick any x with $0 < |x - a| < \delta$. By the triangle inequality:

$$|L_1 - L_2| \leq |L_1 - f(x)| + |f(x) - L_2| < \epsilon + \epsilon = |L_1 - L_2|$$

This gives $|L_1 - L_2| < |L_1 - L_2|$, a contradiction. Hence $L_1 = L_2$.

Exercise 8

Prove that the function $f(x) = x\sin(\frac{1}{x})$ is continuous everywhere, but not differentiable at $x = 0$.

Intermediate Value Theorem

Theorem (Intermediate Value Theorem)

Let I be an interval in $\mathbb{R}$, $f : I \rightarrow \mathbb{R}$ be a continuous function and $a < b$ in I . If $f(a) < f(b)$, and $m \in \mathbb{R}$ is such that $f(a) < m < f(b)$, then there exists $c \in (a, b)$ such that $f(c) = m$.

Basically, if f is continuous, given some $f(a)$ and $f(b)$, f attains all values between $f(a)$ and $f(b)$ for some $x \in (a, b)$

That's just how I Rolle

Theorem (Rolle's Theorem)

Let $f : [a, b] \rightarrow \mathbf{R}$ be a **continuous** and **differentiable** function. Also suppose $f(a) = f(b)$.

Then there exists $c \in (a, b)$ such that $f'(c) = 0$.

It's a Secant? It's a Tangent!? It's LMVT!

Theorem (Lagrange's Mean Value Theorem)

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

If $f : I \rightarrow \mathbb{R}$ is n -times differentiable and $a, x \in I$.
Then, $\exists c$ between a and x such that

$$f(x) = f(a) + (x - a)f'(a) + \cdots + \frac{(x - a)^{n-1}}{(n - 1)!} f^{(n-1)}(a) + R_n(a, x, c)$$

where

$$R_n(x) = \frac{(x - a)^n}{n!} f^{(n)}(c) \quad \text{for some } c \in (a, x).$$

$R_n(x)$ is called the n th Lagrange remainder.

Seriesly useful formulas

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots,$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots,$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots,$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots, \quad |x| < 1.$$

Darboux's Theorem

The derivative of a differentiable function need not necessarily be continuous, but interestingly, it still satisfies the Intermediate Value Property.

Theorem (Darboux's Theorem)

Let $f : (a, b) \rightarrow \mathbb{R}$ be a differentiable function and let α, β with $\alpha < \beta$ be points in (a, b) . If $f'(\alpha) < f'(\beta)$ and $u \in \mathbb{R}$ is such that $f'(\alpha) < u < f'(\beta)$, then there exists $c \in (\alpha, \beta)$ such that $f'(c) = u$.

Exercise 9

Given, $f : [0, 1] \rightarrow [0, 1]$ is a continuous function. Prove that there exists a $c \in [0, 1]$ such that $f(c) = c$.

Exercise 9 - Solution

Special cases: If $f(0) = 0$, take $c = 0$. If $f(1) = 1$, take $c = 1$. Done in either case.

Remaining case: Suppose neither holds. Since $f(0) \in [0, 1]$ and $f(0) \neq 0$, we get $f(0) > 0$. Since $f(1) \in [0, 1]$ and $f(1) \neq 1$, we get $f(1) < 1$.

Define $g(x) = f(x) - x$ on $[0, 1]$.

Then:

$$g(0) = f(0) - 0 = f(0) > 0, \quad g(1) = f(1) - 1 < 0$$

Since f is continuous and x is continuous, g is continuous on $[0, 1]$.

By the Intermediate Value Theorem, since $g(0) > 0$ and $g(1) < 0$, there exists $c \in (0, 1)$ such that

$$g(c) = 0 \implies f(c) - c = 0 \implies f(c) = c.$$

Exercise 10

$$f(x) = x + \frac{1}{x}, \quad a = \frac{1}{2}, \quad b = 2$$

Verify LMVT by finding $c \in (a, b)$ that satisfies $f(b) - f(a) = f'(c)(b - a)$

Exercise 11

Let p and q be two real numbers with $p > 0$. Show that $x^3 + px + q$ has only one real root.

Hint -

To show that at least one real root exists, argue that the polynomial is negative for large negative x , and positive for large positive x , so by **IVT** it must be zero for some x .

To show that only one root exists, assume that two roots exist, which by **Rolle's theorem** implies that for some c , $f'(c) = 0$. Then show that this isn't possible.

Exercise 11

Show that $f(x) = x^4 + 2x^3 - 2$ has exactly one root in $(0,1)$.

Exercise 11

Show that $f(x) = x^4 + 2x^3 - 2$ has exactly one root in $(0,1)$.

Solution -

$$f(1) = 1, \quad f(0) = -2$$

By IVT, at least one root exists in $(0,1)$. Suppose two roots, a and b , exist in $(0,1)$.

By Rolle's Theorem,

$$f'(c) = 0, \quad c \in (a, b)$$

Notice that $f'(x) = 4x^3 + 6x^2 = 0$ has no solutions in the interval $(0,1)$.
Contradiction!!
Hence proved.

Exercise 12 - Inequality

Prove that $|\cos a - \cos b| \leq |a - b|$ for all $a, b \in \mathbf{R}$.

Exercise 12 - Inequality (Use LMVT ;))

Prove that $|\cos a - \cos b| \leq |a - b|$ for all $a, b \in \mathbf{R}$.

Solution -

Let

$$f(x) = \cos x$$

Apply LMVT in $[a, b]$:

$$f'(c) = -\sin c = \frac{\cos a - \cos b}{a - b}$$

Therefore,

$$|\cos a - \cos b| = |\sin c| |a - b| \leq |a - b|$$

Exercise 13 - Inequality again

Prove that

$$\frac{1}{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} \right) + 1 < \sqrt{n} < \frac{1}{2} \left(\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n-1}} \right) + 1$$

for all $n \in \mathbb{N}$.

Exercise 13 - Inequality again (LMVT again)

Let $f(x) = \sqrt{x}$

Apply LMVT in $[n, n+1]$

$$\exists c_n \in (n, n+1) \text{ s.t. } f'(c_n) = \frac{1}{2\sqrt{c_n}} = \sqrt{n+1} - \sqrt{n}$$

$$\frac{1}{2\sqrt{n+1}} < \sqrt{n+1} - \sqrt{n} < \frac{1}{2\sqrt{n}}$$

$$\frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{3}} + \cdots + \frac{1}{2\sqrt{n+1}} < \sqrt{n+1} - 1 < \frac{1}{2\sqrt{1}} + \frac{1}{2\sqrt{2}} + \cdots + \frac{1}{2\sqrt{n}}$$

$$\frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{3}} + \cdots + \frac{1}{2\sqrt{n+1}} + 1 < \sqrt{n+1} < \frac{1}{2\sqrt{1}} + \frac{1}{2\sqrt{2}} + \cdots + \frac{1}{2\sqrt{n}} + 1$$

Hence proved.

Local Maxima and Minima

Let $D \subset \mathbb{R}$, and c be an interior point of D . We say that a function $f : D \rightarrow \mathbb{R}$ has

- a **local minimum** at c if there is $\delta > 0$ such that $(c - \delta, c + \delta) \subset D$ and $f(x) \geq f(c)$ for all $x \in (c - \delta, c + \delta)$.
- a **local maximum** at c if there is $\delta > 0$ such that $(c - \delta, c + \delta) \subset D$ and $f(x) \leq f(c)$ for all $x \in (c - \delta, c + \delta)$.
- a **local extremum** at c if it has a local maximum or a local minimum at c .

Theorem (Necessary condition for local extremum)

Let $D \subset \mathbb{R}$, and let c be an interior point of D . Let $f : D \rightarrow \mathbb{R}$, if f is differentiable at c , and f has a local extremum at c , then $f'(c) = 0$.

Note that this is not a sufficient condition (e.g., $f(x) = x^3$ at $x = 0$)

Sufficient conditions - First derivative test, Second derivative test

Critical Points and Global Extrema

A point $x = c$ is called a **critical point** of f , if either f is differentiable at $x = c$ and $f'(c) = 0$, or f is not differentiable at $x = c$.

Consider a function, $f : [a, b] \rightarrow \mathbb{R}$. Then the global maximum and global minimum of f must occur either at one of the endpoints, or at one of the critical points.

Convex and Concave lens functions :o

A function $f : I \rightarrow \mathbb{R}$ defined on open interval I , is said to be **convex** if given any $x_1 < x < x_2 \in I$,

$$f(x) \leq f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x - x_1).$$

In other words, the chord joining $(x_1, f(x_1))$ and $(x_2, f(x_2))$ lies **above** the graph of f .

Similarly, if the chord joining $(x_1, f(x_1))$ and $(x_2, f(x_2))$ lies **below** the graph of f , for all $x_1, x_2 \in I$, the function is called **concave**.

Identifying convex functions

Theorem

Let $f : I \rightarrow \mathbf{R}$ be twice-differentiable. Then -

$f'' \geq 0$ on I iff f is convex on I .

$f'' \leq 0$ on I iff f is concave on I .

Note that there can be convex/concave functions which are not twice-differentiable, in which case this theorem doesn't apply to such functions. However, our earlier definition still holds in such cases.

A Matter of Inflection

The point $x = c$ is called a **point of inflection** of $f(x)$, if in a small interval $(c - \delta, c)$, $f(x)$ is convex; and in a small interval $(c, c + \delta)$, $f(x)$ is concave, or vice-versa.

Basically, at a point of inflection, the function changes from being convex to concave or vice-versa. The graph crosses its tangent and bends away from the tangent in the other direction.

Theorem (Necessary condition for Point of inflection)

If $f(x)$ is twice-differentiable at $x = c$, and $x = c$ is a point of inflection of $f(x)$, then $f''(c) = 0$.

Exercise 14

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^3 + 3^2, & x \leq 0, \\ (x - 3)^2, & x > 0, \end{cases}$$

Which of the following are true?

- (a) The function $f(x)$ is strictly convex in the interval $(1, 5)$.
- (b) The function $f(x)$ is differentiable at all points in $\mathbb{R}$.
- (c) The point $x = 0$ is a point of inflection for the function $f(x)$.
- (d) The function $f(x)$ has a local maximum at $x = 0$.

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Answer - (a), (c)

Exercise 15

Prove that for $x \in (0, 1)$,

$$e^x < 1 + (e - 1)x$$

using the fact that e^x is convex

Exercise 15 - Solution

Prove that for $x \in (0, 1)$,

$$e^x < 1 + (e - 1)x$$

Hint - Use the fact that e^x is a convex function. Then apply the definition of convex functions in the interval $[0, 1]$.

Exercise 16

If f and g have a point of inflection at $x = 0$, then fg has a point of inflection at $x = 0$. True/False?

Exercise 16 - Solution

If f and g have a point of inflection at $x = 0$, then fg has a point of inflection at $x = 0$. True/False? Let us try to find a counter-example. Let

-

$$f(x) = \begin{cases} x^2, & x < 0 \\ -x^2, & x > 0 \end{cases}$$

$$g(x) = \begin{cases} -x^2, & x < 0 \\ x^2, & x > 0 \end{cases}$$

$$(fg)(x) = -x^4$$

$x = 0$ is not a point of inflection. Hence, the answer is **False**.

Exercise 17

A function is continuously differentiable if it is differentiable, and its derivative is continuous. Which one of the following statements is **FALSE**?

For any continuously differentiable function $g : [0, 1] \rightarrow \mathbb{R}$ with

$$g(0) = 1, \quad g(1) = e, \quad \text{and} \quad g'_+(0) = \frac{1}{2},$$

there is $c \in (0, 1)$ such that

- (a) $g'(c) = 2c g(c)$
- (b) $g'(c) = 1$
- (c) $g'(c) = e - 1$
- (d) $g'(c) = 2e$

Exercise 17 - Solution

Proof of part (c)

Apply LMVT to $g(x)$ in $[0, 1]$

$$\therefore \exists c_1 \text{ s.t. } g'(c_1) = \frac{g(1) - g(0)}{1 - 0} = e - 1$$

Proof of part (b)

Since derivative is continuous,

Apply IVP to $g'(x)$

$$g'(0^+) = 0.5$$

$$g'(1^-) = e - 1 \approx 1.7$$

$$\therefore \exists c_2 \in (0, c_1) \text{ s.t. } g'(c_2) = 1$$

Exercise 17 - Solution

Proof of part (a)

Define

$$f(x) = e^{-x^2}g(x)$$

$$f(0) = f(1) = 1$$

Apply Rolle's Thm in $[0, 1]$

$$\therefore \exists c_3 \in (0, 1) \text{ s.t. } f'(c_3) = 0$$

$$\implies e^{-c_3^2}(g'(c_3) - 2c_3g(c_3)) = 0$$

$$\therefore g'(c_3) = 2c_3g(c_3)$$

Exercise 17 - Solution

A function is continuously differentiable if it is differentiable, and its derivative is continuous. Which one of the following statements is **FALSE**?

For any continuously differentiable function $g : [0, 1] \rightarrow \mathbb{R}$ with

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there is $c \in (0, 1)$ such that

(a) $g'(c) = 2c \, g(c)$

(b) $g'(c) = 1$

(c) $g'(c) = e - 1$

(d) $g'(c) = 2e$

Answer - (d)

Can we go home now?

Not so fast :)

Don't stress too much about grades. How many marks you score in this exam will not affect the number of zeroes in your salary.

So,



(Credits for this slide - Dhairya and Nirav from 2024's TSC :D)

All the best!